ASTRA EVENT
LEARN • PRACTICE • MASTER

RHCSA SA1

SYSTEM ADMIN CHALLENGE

BUILD | CONFIGURE | SECURE | MASTER



COMMAND THE SYSTEM

SECURE EVERY LAYER

MANAGE RESOURCES

MONITOR PERFORMANCE

TROUBLESHOOT LIKE A PRO

REAL SKILLS. REAL EXPERIENCE. REAL ADMIN.

INTRODUCTION

Learn → Practice → Troubleshoot → Document → Submit



This practical event has been designed based on the topics covered in the SA1 rhcsa paper. Therefore, you are expected to have a good understanding of the topics already taught, especially User Management, Group Management, File and Directory Permissions, Service Management, Networking, and Package Installation.

If you have not yet completed the SA1 portion, you can still begin this practical using the knowledge you have acquired so far. However, it is strongly recommended that you study and understand each topic properly before attending this event.

A few tasks included in this practical may go slightly beyond what has been covered in the regular sessions. You are allowed to refer to external resources and documentation to complete such tasks. This is intentional and will help you develop the habit of researching, troubleshooting, and finding solutions independently—an important skill for a Linux System Administrator.

To successfully complete this practical, you must score 70 marks or above. If you do not achieve the required score, you may have to attempt the practical again.

Submission Guidelines

Each participant must document their own work and progress properly. Your submission should include clear screenshots of the tasks you have completed. Wherever possible, you may also include short video recordings or relevant video links as additional evidence of your work. A well-documented submission can help demonstrate your practical knowledge and support your overall score.

Do not submit another person's document, screenshots, or work under your name. Any submission found to contain copied or someone else's work will be rejected. In such a situation, you may also face difficulty getting an opportunity to resubmit.

You do not have to worry if you are unable to complete every task. The purpose of this practical is to understand how much you actually know and how effectively you can apply your knowledge.

Therefore, focus on the tasks you understand, attempt them independently, and demonstrate your own practical skills. It is better to submit your genuine work and knowledge than to submit copied work simply to complete all the tasks.

All the best!

PHASE 1 – OPERATING SYSTEM SETUP & SSH CONNECTIVITY

Step 1: Download and Add the Operating System

Download the Astra Linux operating system file from the Google Drive link provided below and add the operating system to VMware Workstation.

The provided virtual machine contains a Rocky Linux operating system.

Login Credentials:

- Username: root
- Password: 1234

Download Os

https://drive.google.com/file/d/13v4vU-gshg-1j1zfHH_fw3SoaiWBi88Z/view?usp=sharing

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Step 2: Configure the IP Address Manually

After starting the virtual machine, your first task is to configure a suitable static IP address manually.

Before assigning the IP address:

1. Check the network configuration of your main/host operating system.
2. Identify which network and IP address range your host system is using.
3. If your main operating system is Windows, check the Windows IP address, subnet, and default gateway.
4. Select an unused IP address from the same network/subnet for the Astra Linux virtual machine.
5. Use the same default gateway as the Windows host machine.
6. Configure the required network settings manually on the Astra Linux system.
7. Verify that the Astra Linux system has proper network connectivity.

Important: Do not simply copy the Windows IP address. The IP address assigned to the AstraLinux system must be a different, unused IP address within the same network/subnet.

Step 3: Configure SSH

After configuring the network, install and configure the OpenSSH/SSH service on the Rocky Linux system.

If you face difficulty installing the required packages, you may use the following repositories:

BaseOS:
https://download.rockylinux.org/pub/rocky/9/BaseOS/x86_64/os/

AppStream:
https://download.rockylinux.org/pub/rocky/9/AppStream/x86_64/os/

EPEL:
https://dl.fedoraproject.org/pub/epel/9/Everything/x86_64/

Configure the required repositories only if they are necessary for completing the installation.

After installing and configuring SSH:

- Make sure the SSH service is running.
- Ensure that the service starts automatically when required.
- Verify that the required SSH port is accessible.
- Verify network connectivity between the host Windows system and the Astra Linux virtual machine.

Step 4: Connect Using PuTTY

From the Windows host machine, use PuTTY to connect to the Astra Linux virtual machine through SSH.

You must successfully establish an SSH connection from:

Windows Host → PuTTY → Astra Linux

Verify that you can log in successfully through SSH.

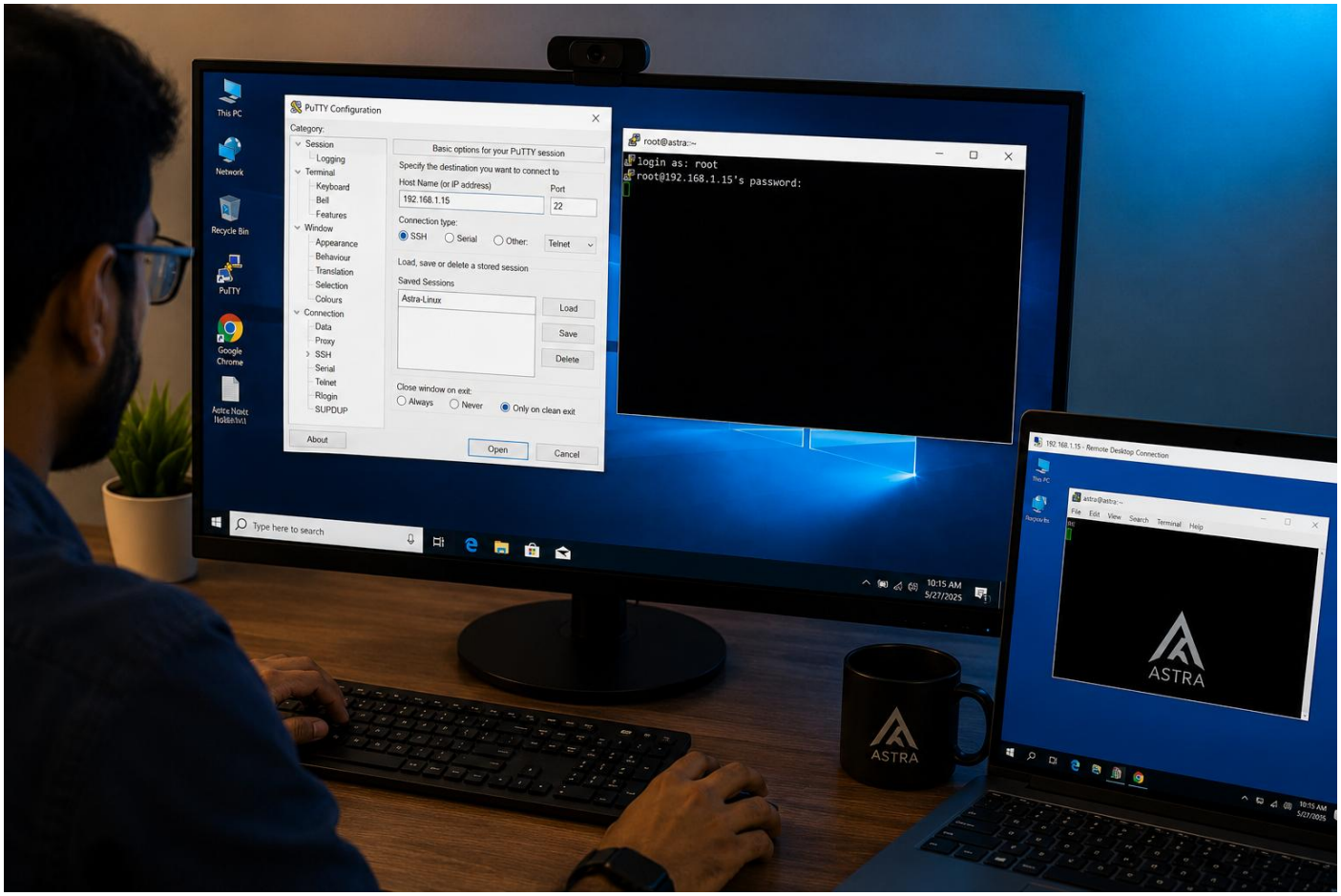
Final Verification for Phase 1

Before moving to Phase 2, make sure the following are working correctly:

- Astra Linux virtual machine is running successfully.
- A valid static IP address has been configured manually.

- The IP address belongs to the correct network/subnet.
- The correct default gateway has been configured.
- Network connectivity between the Windows host and Astra Linux is working.
- SSH has been installed and configured.
- The SSH service is running.
- PuTTY can successfully connect to the Astra Linux system.
- SSH login is working correctly.

Note: You are expected to troubleshoot and resolve any issues you encounter during this phase. Do not depend entirely on step-by-step instructions. Use the knowledge you have gained from the course and refer to appropriate documentation when required.



PHASE 2 – USER MANAGEMENT, PERMISSIONS, MONITORING & GRAPHICAL ENVIRONMENT

In this phase, you will perform user management, privilege configuration, directory permission management, system monitoring, and graphical environment configuration on the Astra operating system.

Task 1 – Create and Configure Your User Account

- Create a new user account using your own name.
- The user account must have sudo privileges without requiring a password.
- This user account has been created specifically for this practical event. Therefore, configure the account to automatically expire on December 31, 2026.

You must verify that:

- The user has been created successfully.
- The user can use sudo without entering a password.
- The account expiry date is correctly configured as December 31, 2026.

Task 2 – Create and Configure the /astra Directory

Create a directory named:

/astra

- The directory must be configured so that any user can create files or directories inside it, but only the user who owns a particular file or directory can delete it.
- Configure the appropriate permissions to achieve this requirement.
- You must verify the permissions and demonstrate that the required behavior works correctly.

Task 3 – Monitoring and Activity Logging

All relevant activities taking place on the system must be monitored for troubleshooting purposes.

Create a file named:

monitoring

- **This file will be used to store the required monitoring/log information generated by the activities performed on the system.**
- **Configure the appropriate logging mechanism so that relevant messages and log information can be collected and stored for later investigation.**
- **The purpose of this requirement is to ensure that, if any issue occurs during the practical, the monitoring information can be reviewed to identify and troubleshoot the problem.**
- **You must demonstrate that the required log information is being recorded correctly.**

Task 4 – Configure a Graphical Environment

The Astra operating system is currently configured to work primarily through a Command Line Interface (CLI).

Your task is to configure a suitable Graphical User Interface (GUI) for the operating system.

After completing the configuration, verify that:

- **The required graphical packages are installed.**
- **The graphical environment starts successfully.**
- **The system can be used through the GUI.**
- **The required graphical services are running correctly.**

Task 5 – Remote Desktop Access

After successfully configuring the graphical environment, investigate and configure a suitable Remote Desktop solution.

The objective is to determine whether the Astra operating system can be accessed and used graphically from the external/host system through a remote desktop connection.

You must:

1. Configure the required remote desktop components.
2. Start and enable the required services.
3. Configure the necessary network/firewall settings.
4. Attempt a remote graphical connection from the host machine.
5. Verify that the Astra graphical desktop can be accessed successfully.

If you encounter any problems, troubleshoot them using the appropriate system logs, service status, network checks, and configuration files.

Phase 2 Completion Criteria

Phase 2 will be considered complete when you can demonstrate:

- Your own user account has been created.
- Sudo access works without a password.
- The account expires on December 31, 2026.
- /astra has the required special permission behavior.
- The monitoring file has been created.
- Required system activity/log information is being collected.
- A graphical environment has been successfully configured.
- The Astra desktop can be accessed graphically.
- Remote Desktop access has been tested successfully.
- Any issues encountered have been investigated and documented.

Important: You are encouraged to research the required commands and configuration methods when necessary. The objective is not simply to execute commands, but to understand why each configuration is required and how to troubleshoot it when something goes wrong.

SUBMISSION GUIDELINES

Your submitted document must clearly explain all the steps you personally performed during the practical. Every major task you completed should be documented in a clear and organized manner, and screenshots are mandatory as evidence of your work. Good technical documentation should be clear, consistent, and easy for another person to understand.

The purpose of this document is not to provide copied theory or ready-made notes. You must explain what you understood, what you did, how you performed each step, and what result you obtained.

Your document must include:

- The steps you performed for each task.
- Relevant screenshots showing your actual work.
- A brief explanation of what you did and why.
- The result or verification of each completed task.
- Any problems or errors you encountered.
- The troubleshooting steps you followed to resolve those problems.
- Relevant log information or evidence wherever applicable.

Screenshots must support your explanation. Do not simply insert screenshots without explaining what they demonstrate; a screenshot is most useful when it has clear context and a short description.

IMPORTANT

- Do not submit another person's work.
- Documents written by someone else, copied material, or documents consisting only of AI-generated notes from ChatGPT or other AI tools will be rejected.
- You may refer to external resources when you need help understanding a task, but your final document must reflect your own understanding and your own practical work.

The objective is very simple:

Anyone who reads your document should be able to clearly understand what you did, how you did it, and what you learned from the process.

Therefore, prepare a clean, well-organized, and technically accurate document containing your own steps, screenshots, explanations, and verification results.

TOTAL MARKS & OPTIONAL EXPERT CHALLENGE

Total Assessment – 100 Marks

Phase 1 and Phase 2 together carry a total of 100 marks.

Your final score will be calculated based on your performance in both phases.

